

Hybrid Automation Experiences – Communication, Coordination, and Collaboration within Human-AI Teams

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Abstract

Automated systems and AI-assisted workflows are evolving from support tools to sophisticated collaborators in so-called “hybrid” teams with human and AI-based members. This evolution presents new challenges and opportunities in the field of “Automation Experience”, particularly in how these team members communicate, coordinate, and collaborate. This workshop aims to contribute to the design of automated systems that augment humans’ capabilities, maintain human autonomy, and create positive automation experiences in human-AI teams. Through presentations, a keynote talk, discussions, and interactive collaborative activities, researchers and practitioners from different fields will address challenges regarding communication, coordination, and collaboration within hybrid human-AI teams. The workshop aims to spark innovative research directions and foster collaborative interdisciplinary initiatives that will advance our understanding of automation experiences in an increasingly AI-dominated landscape.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Automation, user experience, artificial intelligence, hybrid teams, human-AI teaming

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1 Motivation and Goals

Over the last few years, automated systems have been becoming integral to various facets of society, from individual households to corporate environments and government institutions. As these systems proliferate and become accessible to a broader audience, investigating the users’ “Automation Experience” is critical to understanding such systems’ requirements, design principles, and user experience factors.

During this expansion, we are witnessing significant advancements in Artificial Intelligence (AI) technologies [7]. The integration of AI into automated systems enhances their capabilities, allowing them to perform more complex tasks that were previously often reserved for humans. For example, with the development of state-of-the-art (generative) AI, these systems now possess sophisticated creative and intellectual capabilities (e.g., drawing, coding, idea generation, planning, etc.). These recent developments allow for a more dynamic and enhanced interplay of humans and (semi-)automated systems forming “hybrid teams” where human and AI-based members cooperate to achieve common goals.

To ensure effective, meaningful, and sound teamwork and thus fully exploit the potential of such hybrid teams, understanding the dynamics and mechanisms between humans and AI-based automated systems is paramount. In addition to obvious benefits such as increased efficiency and reduced workload, hybrid teams carry

risks, such as an over-reliance on automated systems and the need to ensure that they complement users rather than replace their skills [19]. Furthermore, the integration of such a sophisticated level of automation also has the potential to mislead users [15]. Thus, studying the collaboration of these hybrid teams to draw insights into their effective and reliable deployment is crucial.

Recent research on automation experiences with single appliances has highlighted challenges such as system intelligibility and user engagement [2, 3, 9]. With the rise of AI, increased capabilities of automated systems, and more complex interaction between humans and AI-based systems, new challenges emerge. In our workshop, we set out to extend the concept of automation experience to hybrid human-AI teams and work toward creating positive “hybrid automation experiences” in an interdisciplinary and cross-domain approach. Based on these considerations, the workshop addresses three core challenges for hybrid human-AI teams: communication, coordination, and collaboration.

The objectives of the workshop are as follows:

- Facilitate the exchange of recent research findings, methodologies, and advancements related to hybrid automation experiences,
- Discuss major challenges regarding the communication, coordination, and collaboration in hybrid human-AI teams,
- Identify promising future research topics in the form of project ideas and a research agenda,
- Expand and strengthen a multi-disciplinary network of automation experience researchers.

2 Challenges

In the workshop, we will elaborate on the following three core challenges within human-AI teams.

2.1 Communication in Hybrid Teams

Effective communication, a cornerstone of collaboration, is essential to achieve success and productivity in teamwork [17]. Recent studies have identified several challenges in communication within hybrid human-AI teams (cf. [11]), which are becoming increasingly relevant as more organizations integrate AI technologies and, at the same time, the size of these teams grows, both in terms of human and AI participants.

Human teams generally communicate more frequently than hybrid human-AI teams, particularly in the early stages of work processes [23]. However, when unexpected changes or constraints arise, communication within hybrid teams can surge, sometimes tripling in volume. This indicates that hybrid teams may face difficulties in maintaining consistent communication patterns. Adapting to dynamic situations while maintaining effective communication presents a key challenge for such teams.

For human members of hybrid teams, seamless and natural communication is highly desirable. However, achieving a high degree of natural interaction remains a significant hurdle, especially when dealing with uncertainty [16]. Although recent advances in generative AI, particularly large language models (LLMs), offer promising new methods for team communication, numerous open questions and challenges persist concerning the design and user experience with these conversational systems (cf. [26]).

Lastly, for communication within a team to remain efficient, the development and maintenance of shared understanding and situation awareness are essential (cf. [1, 6]). This raises questions about how to verify the accurate comprehension of communication through feedback among team members and how to ensure that understanding of tasks and individual needs remains current [11].

Research questions regarding the communication within hybrid teams include:

- How to exploit new modalities to communicate naturally and interact effectively within hybrid teams?
- What effects do different communication patterns have on the experience of human group members?
- How to create and maintain a shared understanding within hybrid teams?

2.2 Coordination in Hybrid Teams

Recent advancements in AI have enabled automated systems to perform tasks traditionally done by human knowledge workers. These tasks range from high-level cognitive processes, including decision-making and creative ideation [20], to more specific technical activities, like data analysis and programming [10, 14]. This shift in the field raises important questions about how to allocate tasks in hybrid teams.

The orchestration of these hybrid teams presents several challenges [4]. An effective task distribution between humans and automated systems requires a detailed understanding of the capabilities of each partner [21], particularly as AI systems rapidly advance in the handling of complex tasks. This dynamic environment requires a continuous reassessment of allocation strategies and real-time evaluation of both AI and human worker capabilities and satisfaction [13]. Crucially, this orchestration must balance the drive for efficiency through AI integration with the imperative to maintain meaningful human oversight and intervention [25]. This balance aims to leverage the complementary strengths of both human and AI team members while preserving human agency, critical thinking skills, and task satisfaction [12]. The complexity of these interrelated factors underscores the need for adaptive coordination mechanisms. Consequently, important research questions regarding the coordination of human-AI teams include:

- How to design adaptive coordination mechanisms for human-AI teams that balance the distribution of task instances, considering real-time performance, availability, and the unique capabilities of both human and automated partners?
- In what ways does the dynamic allocation of tasks between humans and AI systems influence user satisfaction and overall quality of automation experiences?
- What strategies can create engaging automation experiences that maintain meaningful human involvement and ensure appropriate oversight in AI-assisted processes?

2.3 Collaboration in Hybrid Teams

Automated systems have traditionally been used to assist users in predefined tasks, often operating in isolation with limited collaboration between users and the systems. However, with the rise of AI, the role of automated systems is shifting from simple support tools to sophisticated team members. Yet, humans and automated

systems possess different strengths and weaknesses and, thus, are complementary to each other [21]. This evolution mirrors challenges for creating systems that facilitate collaboration taking into account individuals' needs, traits, and preferences rather than designing for a "one-size-fits-all" solution.

Additionally, automation has traditionally been seen as a way to boost productivity and improve product quality by either assisting operators or fully taking over tasks [24]. However, this has led to issues like deskilling [18], complacency, and over-reliance on automation, especially with the rise of AI technology, which has even impacted unexpected areas like dataset production [5]. While some critical domains remain unaffected, others, such as translation, image, and text production, are fully influenced by AI, shifting users' roles from doing tasks to supervising them. AI has made automation more complex and dynamic, often blurring the line between user control and automation, as seen in tools like translation software.

Furthermore, as automation systems become more autonomous, users are increasingly relying on these systems to make decisions, generate solutions, and even take initiative in collaborative environments. This reliance introduces challenges around effective collaboration: users must be able to properly comprehend automated systems' advice while also being able to intervene or override the system when necessary. Over time, the dynamics of this collaboration may change as human team members become more accustomed to the capabilities of automated systems and may start to over-rely on them [19], potentially diminishing human oversight and critical thinking. Moreover, long-term collaboration raises questions about how these systems evolve alongside human partners—will automated systems learn and adapt to human workflows, or will they inadvertently shape human behavior to align with their own processes? At the same time, users also advance their capabilities through interaction with these systems [22]. Thus, it is critical to understand how users adapt to the enhanced capabilities of automated systems.

Consequently, the following research questions in this area include:

- How to effectively design collaboration scenarios that leverage the rising capabilities of automated systems, ensuring a pleasant automation experience?
- How does the nature of collaboration between users and automated systems evolve with technological advancements?
- What strategies can ensure the reliable integration of automated systems into human workflows, particularly as human reliance on automated systems develops over time?

3 Organizers

The members of the organizing team bring extensive expertise across the full range of user-centered automation and AI research. They have successfully hosted both in-person, hybrid, and virtual workshops on various topics, including user experience, automation in daily life, transportation, and safety-critical environments (e.g., CHI, MobileHCI, AutomotiveUI). Team members have significant experience chairing workshops and conferences (e.g., MobileHCI, HRI, AutomotiveUI, Persuasive), further highlighting their leadership and knowledge in the field.

- **Philipp Spitzer** (main contact) is a research lab lead in Information Systems at the Karlsruhe Institute of Technology. He leads the Applied AI in Service Systems lab and conducts research projects on applied machine learning topics. His main focus is on the interplay of knowledge and human-AI interaction, focusing on designing robust and effective collaborative systems.
- **Matthias Baldauf** is a Professor for Business Informatics at the Eastern Switzerland University of Applied Sciences in St.Gallen. He leads user-centered projects in the domain of human-automation interaction in smart manufacturing and office workplaces.
- **Philippe Palanque** is a Professor of Computer Science at the University of Toulouse III. His research focuses on interactive systems design, development, certification, and deployment in various safety critical contexts (e.g., aircraft cockpits, satellite workstations).
- **Virpi Roto** is a Vice Dean of Impact at Aalto University. She studies the means to design automation that improves user experience, for example, in maritime and industrial contexts.
- **Katelyn Morrison** is a PhD student at the Human-Computer Interaction Institute at Carnegie Mellon University and focuses her research on improving human-AI collaboration.
- **Domenique Zipperling** is a PhD student at the Chair of Information Systems and Human-Centric Artificial Intelligence at the University of Bayreuth and a research assistant at the Business & Information Systems Engineering branch of Fraunhofer FIT. His research primarily focuses on the use of synthetic data in human-AI interaction.
- **Joshua Holstein** is a PhD student at the Digital Service Innovation chair at the Karlsruhe Institute of Technology. His main work focuses on enhancing human-AI interaction by improving the user's understanding of the AI systems underlying data and studying how these improvements impact user reliance and performance in human-AI decision-making.

4 Website

The website <https://matthiasbaldauf.com/automationxp25> will provide the workshop description and objectives, a call for papers with suggested topics, a detailed workshop schedule, opportunities for participation, and information about the organizers.

5 Pre-Workshop Plans

The workshop will be publicized via prominent HCI-related mailing lists (such as CHI Announcements, Ubicomp Announcements, etc.) and appropriate websites. In addition, personal invitations will be sent to our scientific network, including past participants and organizers of similar workshops. All promotional materials will include a link to the workshop website. We will invite the submission of position papers, up to 6 pages in length (excluding references), following the CEUR-ART format. These papers should outline the participants' contributions to the workshop, with acceptable submission types including work-in-progress, concrete research ideas, novel perspectives, and demos addressing research questions relevant to the outlined or related challenges in automation engagement.

The organizing committee will review and select submissions based on their relevance to the workshop's theme, originality, significance, and quality. We aim to accept about 25 submissions, which will be published on the workshop website prior to the event. In addition, the authors of the accepted papers will create virtual posters that will be made available online (see 8.1.2).

6 Accessibility

Authors of accepted position papers will be encouraged to ensure their papers are accessible. During the preparation of the final camera-ready versions, organizers will provide authors with guidance on document tagging, adding alternative text for images, setting tab orders, and more. Additionally, to enhance the workshop's accessibility for individuals with disabilities, we will consider offering real-time subtitles and/or transcripts for the video conference, depending on participants' needs.

7 Asynchronous Engagement

Asynchronous participation for individuals facing technical or accessibility challenges will be facilitated through various activities both before and after the workshop. One week prior to the event, participants' position papers will be made available on the workshop website. Additionally, a link to a Miro board featuring virtual posters will be provided, enabling participants to familiarize themselves with all accepted contributions in advance. This approach allows those with limited English proficiency or restricted internet bandwidth to prepare and engage ahead of time. Following the workshop, all materials generated during the event will be published on the website for asynchronous access.

8 Workshop Format and Structure

In designing the format and structure of the workshop, we have incorporated insights and best practices gained from previous related workshops. Our workshop is designed as a one-day event, lasting approximately six hours. It will follow a hybrid format, welcoming both in-person and remote participants through Zoom, supported by Jabra speakers and an external webcam for enhanced interaction. The schedule has been tailored to this hybrid approach, incorporating longer breaks than those typically found in solely in-person workshops to ensure smooth participation for all attendees. Table 1 shows an overview of the preliminary workshop schedule.

8.1 Morning Session

The morning session of the workshop comprises a keynote, poster presentations, and discussions.

8.1.1 Introduction and Keynote. The workshop will begin with an official welcome to the participants, during which the organizers will present the theme of the workshop and outline its main objectives. This will be followed by brief introductions from the organizers and a round of participant introductions. A keynote talk will then be delivered by Professor Albrecht Schmidt, Professor for Human-Centered Ubiquitous Media at LMU Munich, who will provide an overview of recent research on Human-AI collaboration.

8.1.2 Poster Madness. Participants will be required to create concise virtual posters summarizing their workshop contributions,

which will be displayed on a shared digital "Miro" board. Prior to the workshop, the organizers will categorize and arrange these posters based on thematic relevance. During two dedicated "poster madness" sessions, participants will present their recent research using these posters. Depending on the total number of participants, presentation times will be limited to 5 to 8 minutes. This format was intentionally chosen to promote fast-paced visual presentations, facilitate easy reference to participants' work in subsequent sessions, and provide an interactive medium for contributions. Additionally, participants will be able to leave comments, questions, and suggestions in the form of virtual post-its attached to the posters.

We will prepare a shared document for joint note-taking during the presentations by both the in-person and remote participants. These notes are supposed to provide inspiration for the following group work and the advancement of the position papers by the authors.

8.2 Afternoon Session

The afternoon session will include a series of collaborative group activities, followed by a presentation of the outcomes and a subsequent plenary discussion. The session will conclude with a summary provided by the organizers.

8.2.1 Eliciting Themes and Ideation. After participants have introduced and discussed their recent relevant work, they will identify common themes, research questions, and methodological approaches for designing, assessing, and measuring engaging automation experiences. This process will be facilitated by the organizers using a collective Miro board, leveraging its polling features. To spark the discussion, the organizers will provide a few example themes to initiate the idea-generation phase.

Group formation will be based on participants' locations to ensure efficient collaboration. In-person attendees will be grouped together at the conference venue, while remote participants will engage in Zoom breakout rooms. Physical materials, such as paper sheets, colored pens, and post-its, will be provided for the in-person groups, while additional digital whiteboards will be prepared for remote collaboration.

To promote cross-group knowledge exchange, we will implement a "World Café"-style phase, encouraging participants to visit other groups—either by physically joining in-person groups or virtually participating in remote groups via Zoom—to discuss interim findings. The outcomes of these group activities may include new research ideas and draft proposals for potential (joint) research projects.

8.2.2 Group Presentations and Wrap-Up. Towards the end of the workshop, both the on-site and remote groups will reconvene in a plenary session to present the outcomes of their discussions. All participants are encouraged to provide feedback on the research concepts and collaborate to refine and enhance the proposed plans. The organizers will then conclude the session by outlining a roadmap for future research and initiatives based on the ideas generated. As part of this process, efforts will be made to identify relevant research communities, timelines, funding opportunities, and project formats. Additionally, plans for a dedicated programmatic publication will be developed at this stage.

Table 1: Preliminary schedule for a hybrid workshop format.

Time	Phase (details in text)	Methods and tools for hybrid setup
08:30 - 09:00	<i>Arrival of participants</i>	Preparation for tech troubleshooting
09:00 - 09:15	<i>Welcome and introduction</i>	Video conferencing/streaming in plenary
09:15 - 09:35	<i>Participants introduction</i>	Video conferencing/streaming in plenary
09:35 - 09:55	<i>Keynote by Albrecht Schmidt</i>	Video conferencing/streaming in plenary
09:55 - 10:10	<i>Coffee break</i>	Virtual break-out room for (off-topic) chat
10:10 - 11:00	<i>Poster madness 1</i>	Video conferencing/shared virtual poster boards
11:00 - 11:10	<i>Coffee break</i>	Virtual break-out room for (off-topic) chat
11:10 - 12:00	<i>Poster madness 2</i>	Video conferencing/shared virtual poster boards
12:00 - 13:30	<i>Lunch Break</i>	-
13:30 - 13:50	<i>Eliciting Themes</i>	Hybrid discussion, polling tool, forming of smaller groups
13:50 - 14:35	<i>Ideation Session</i>	On-site group work (posters), virtual break-out rooms (collaborative whiteboards),
14:35 - 15:00	<i>Hybrid world-cafe</i>	Switching groups (on-site and remote) for discussion
15:00 - 15:20	<i>Coffee break</i>	Virtual break-out room for (off-topic) chat
15:20 - 16:00	<i>Group presentations</i>	Video conferencing/streaming in plenary
16:00 - 16:30	<i>Future work and wrap-up</i>	Video conferencing/streaming in plenary

9 Post-Workshop Plans

The workshop documentation and outcomes will be made accessible on the official workshop website. This will include photos, screenshots from the interactive sessions, both analog and digital posters, as well as plans for future initiatives. To enhance visibility and ensure long-term preservation, we plan to publish the workshop proceedings on CEUR-WS.

Furthermore, we will contact potential venues for a special journal issue or a magazine article on the workshop contributions and outcomes. One example is the journal on “Personal and Ubiquitous Computing,” where several workshop organizers edited a theme issue on “Everyday Automation Experience” [8]. In addition, together with interested participants, the organizers will discuss opportunities for related workshops at suitable conferences to extend the discussions of this workshop (e.g., MobileHCI, UbiComp).

10 Call for Participation

Automated systems have become an essential component across diverse sectors, ranging from individual households to large-scale organizations. As Artificial Intelligence (AI) and generative technologies advance rapidly, these systems are transitioning from mere support tools to active, sophisticated collaborators within “hybrid” teams comprising both human and AI members. This workshop aims to delve into the multifaceted dynamics of the hybrid automation experience, emphasizing critical issues related to communication, coordination, and collaboration in such teams.

We invite participants to submit position papers outlining their recent or forthcoming research in this area and to engage in meaningful discussions on designing automation systems that enhance user autonomy, prevent over-dependence, and foster productive collaborative outcomes.

Relevant topics of interest include, but are not limited to:

- Effective communication strategies in hybrid human-AI teams
- Coordination and task allocation between human and AI participants
- Methods for coping with the collaborative dynamics between users and AI systems
- Ensuring user autonomy and critical thinking while working with advanced AI systems
- Addressing the risks of over-reliance on AI-based automation
- Designing automation that complements human capabilities and maintains meaningful engagement
- Designing intuitive user interfaces for seamless interaction with automated systems in hybrid teams
- Enhancing user adaptability to evolving AI system capabilities over time

Papers must be formatted according to the one-column CEUR-ART style and comprise up to 6 pages (excl. references) and submitted in PDF format (non-anonymized) to <https://easychair.org/conferences/?conf=automationxp25>. The submissions will be reviewed by the organizers based on relevance, originality, significance, and quality. Upon acceptance, at least one author of each accepted paper must attend the workshop.

Important dates:

- Position paper deadline: February 25th, 2025
- Acceptance notification: March 1st, 2025
- Camera-ready versions due: April 9th, 2025
- Workshop date: April 27th, 2025

Website: <https://matthiasbaldauf.com/automationxp25>

11 Expected size of attendance

We expect about 25 registered attendees, with at least 15 of them participating on-site.

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